

Equations and Inequalities

1 Linear

1. $3x + 7 = 22$
2. $4 - 2x = 18$
3. $7x + 5 = 3x + 25$
4. $3x + 5 > 20$
5. $4x - 7 \leq 17$
6. $6 - 2x < 18$

2 Brackets

1. $3(x + 4) = 27$
2. $4(2x - 3) = 3x + 18$
3. $2(3x + 5) - 4 = 4x + 16$
4. $3(x + 4) > 21$
5. $4(2x - 3) \geq 3x + 12$
6. $5(x - 3) + 2(x + 1) \geq 30$

3 Fractions

1. $\frac{x}{3} = 7$
2. $\frac{x}{3} + \frac{x}{2} = 10$
3. $\frac{2x+1}{3} - \frac{x-2}{4} = 5$
4. $\frac{x}{4} + 3 \leq 8$
5. $\frac{x}{3} + \frac{x}{2} < 10$
6. $\frac{2x+1}{3} - \frac{x-2}{4} \leq 5$

4 Algebraic Problem Solving

For each problem, define a variable, construct an equation or inequality, and solve it algebraically.

1. A number increased by 7 is equal to 25. Find the number.
2. Three times a number decreased by 8 is equal to 19. Find the number.
3. The sum of two consecutive integers is 57. Find the two integers.
4. A rectangle has a perimeter of 54 cm. Its length is 3 cm greater than its width. Find its dimensions.
5. A taxi charges a fixed fee of \$5 plus \$2.50 per kilometre. A passenger pays \$32.50. How many kilometres did the passenger travel?

5 Challenge

Solve the following equations and inequalities algebraically.

1. $\frac{3x-2}{4} + \frac{x+1}{2} = 9$
2. $\frac{2(x+3)}{5} - \frac{x-1}{3} = 4$
3. $\frac{3x-2}{4} + \frac{x+1}{2} > 9$
4. $\frac{2(x+3)}{5} - \frac{x-1}{3} \geq 4$

6 Hard Challenge

Solve the following problems algebraically.

1. Determine all values of a for which the equation $(a - 1)x + 2a = 3x + 5$, has a solution satisfying $x > 1$.
2. Find all positive integers x and y satisfying $x + y = 20$ and x divides y .
3. Find all positive integers x and y satisfying $xy - 2x - 3y = 10$.
4. Solve over the real numbers
$$\frac{x+1}{x-1} + \frac{x-1}{x+1} = 3$$
5. Which sets of three consecutive integers have a sum equal to their product? Prove that these are the only such sets.
6. Determine all values of k for which $|x - 1| + |x + 1| = k$ has exactly two solutions.

1 Linear

1. $3x + 7 = 22$

$$3x = 15$$

$$x = 5$$

2. $4 - 2x = 18$

$$-2x = 14$$

$$x = -7$$

3. $7x + 5 = 3x + 25$

$$4x = 20$$

$$x = 5$$

4. $3x + 5 > 20$

$$3x > 15$$

$$x > 5$$

5. $4x - 7 \leq 17$

$$4x \leq 24$$

$$x \leq 6$$

6. $6 - 2x < 18$

$$-2x < 12$$

Dividing by -2 reverses the inequality:

$$x > -6$$

2 Brackets

1. $3(x + 4) = 27$

$$x + 4 = 9$$

$$x = 5$$

2. $4(2x - 3) = 3x + 18$

$$8x - 12 = 3x + 18$$

$$5x = 30$$

$$x = 6$$

3. $2(3x + 5) - 4 = 4x + 16$

$$6x + 10 - 4 = 4x + 16$$

$$6x + 6 = 4x + 16$$

$$2x = 10$$

$$x = 5$$

4. $3(x + 4) > 21$

$$x + 4 > 7$$

$$x > 3$$

5. $4(2x - 3) \geq 3x + 12$

$$8x - 12 \geq 3x + 12$$

$$5x \geq 24$$

$$x \geq \frac{24}{5}$$

6. $5(x - 3) + 2(x + 1) \geq 30$

$$5x - 15 + 2x + 2 \geq 30$$

$$7x - 13 \geq 30$$

$$7x \geq 43$$

$$x \geq \frac{43}{7}$$

3 Fractions

1. $\frac{x}{3} = 7$

$$x = 21$$

2. $\frac{x}{3} + \frac{x}{2} = 10$

Multiplying by 6:

$$2x + 3x = 60$$

$$5x = 60$$

$$x = 12$$

3. $\frac{2x+1}{3} - \frac{x-2}{4} = 5$

Multiplying by 12:

$$4(2x + 1) - 3(x - 2) = 60$$

$$8x + 4 - 3x + 6 = 60$$

$$5x + 10 = 60$$

$$5x = 50$$

$$x = 10$$

4. $\frac{x}{4} + 3 \leq 8$

$$\frac{x}{4} \leq 5$$

$$x \leq 20$$

5. $\frac{x}{3} + \frac{x}{2} < 10$

Multiplying by 6:

$$2x + 3x < 60$$

$$5x < 60$$

$$x < 12$$

6. $\frac{2x+1}{3} - \frac{x-2}{4} \leq 5$

Multiplying by 12:

$$4(2x + 1) - 3(x - 2) \leq 60$$

$$8x + 4 - 3x + 6 \leq 60$$

$$5x + 10 \leq 60$$

$$5x \leq 50$$

$$x \leq 10$$

4 Algebraic Problem Solving

For each problem, define a variable, construct an equation or inequality, and solve it algebraically.

1. A number increased by 7 is equal to 25. Find the number.

Let x be the number.

$$x + 7 = 25 \longrightarrow x = 18$$

Therefore, the number is 18.

2. Three times a number decreased by 8 is equal to 19. Find the number.

Let x be the number.

$$3x - 8 = 19 \longrightarrow 3x = 27 \longrightarrow x = 9$$

Therefore, the number is 9.

3. The sum of two consecutive integers is 57. Find the two integers.

Let x be the first integer. The next integer is $x + 1$.

$$x + (x + 1) = 57 \longrightarrow 2x + 1 = 57 \longrightarrow 2x = 56 \longrightarrow x = 28$$

The two integers are 28 and 29.

4. A rectangle has a perimeter of 54 cm. Its length is 3 cm greater than its width. Find its dimensions.

Let x be the width. Then the length is $x + 3$.

$$2x + 2(x + 3) = 54 \longrightarrow 4x + 6 = 54 \longrightarrow 4x = 48 \longrightarrow x = 12$$

The width is 12 cm and the length is

$$12 + 3 = 15 \text{ cm.}$$

Therefore, the dimensions are 12 cm by 15 cm.

5. A taxi charges a fixed fee of \$5 plus \$2.50 per kilometre. A passenger pays \$32.50. How many kilometres did the passenger travel?

Let x be the number of kilometres travelled.

$$5 + 2.50x = 32.50 \longrightarrow 2.50x = 27.50 \longrightarrow x = 11$$

Therefore, the passenger travelled 11 km.

5 Challenge

Solve the following equations and inequalities algebraically.

1. $\frac{3x-2}{4} + \frac{x+1}{2} = 9$

Multiplying by 4:

$$3x - 2 + 2(x + 1) = 36$$

$$3x - 2 + 2x + 2 = 36$$

$$5x = 36$$

$$x = \frac{36}{5}$$

2. $\frac{2(x+3)}{5} - \frac{x-1}{3} = 4$

Multiplying by 15:

$$6(x + 3) - 5(x - 1) = 60$$

$$6x + 18 - 5x + 5 = 60$$

$$x + 23 = 60$$

$$x = 37$$

3. $\frac{3x-2}{4} + \frac{x+1}{2} > 9$

Multiplying by 4:

$$3x - 2 + 2(x + 1) > 36$$

$$3x - 2 + 2x + 2 > 36$$

$$5x > 36$$

$$x > \frac{36}{5}$$

4. $\frac{2(x+3)}{5} - \frac{x-1}{3} \geq 4$

Multiplying by 15:

$$6(x + 3) - 5(x - 1) \geq 60$$

$$6x + 18 - 5x + 5 \geq 60$$

$$x + 23 \geq 60$$

$$x \geq 37$$

6 Hard Challenge

1. Determine all values of a for which the equation $(a-1)x + 2a = 3x + 5$ has a solution satisfying $x > 1$.

Solution.

$$(a-1)x + 2a = 3x + 5 \longrightarrow (a-4)x = 5 - 2a \longrightarrow x = \frac{5-2a}{a-4}.$$

For $x > 1$,

$$\frac{5-2a}{a-4} > 1 \longrightarrow \frac{5-2a-(a-4)}{a-4} > 0 \longrightarrow \frac{9-3a}{a-4} > 0 \longrightarrow \frac{3-a}{a-4} > 0.$$

The critical values are $a = 3$ and $a = 4$. Since $a = 4$ is excluded, we analyze the signs:

a	$(-\infty, 3)$	$(3, 4)$	$(4, \infty)$
$3-a$	+	-	-
$a-4$	-	-	+
$\frac{3-a}{a-4}$	-	+	-

Therefore, $\frac{3-a}{a-4} > 0 \longrightarrow 3 < a < 4$.

2. Find all positive integers x and y satisfying $x + y = 20$ and x divides y .

Solution.

Since $x \mid y$, y is a positive integer multiple of x . Therefore, there exists $n \in \mathbb{Z}^+$ such that $y = nx$. Substituting $y = nx$ into $x + y = 20$ gives

$$x + y = 20 \longrightarrow x + nx = 20 \longrightarrow x(n+1) = 20.$$

Since x and $n+1$ are positive integers and their product is 20, both x and $n+1$ must be positive divisors of 20. The positive factor pairs of 20 are

$$1 \cdot 20, \quad 2 \cdot 10, \quad 4 \cdot 5, \quad 5 \cdot 4, \quad 10 \cdot 2, \quad 20 \cdot 1. \longrightarrow n+1 \in \{2, 4, 5, 10, 20\},$$

Then,

$$n \in \{1, 3, 4, 9, 19\}.$$

For each value of n , we use

$$x = \frac{20}{n+1} \quad \text{and} \quad y = nx.$$

n	$n+1$	$x = \frac{20}{n+1}$	$y = nx$
1	2	10	10
3	4	5	15
4	5	4	16
9	10	2	18
19	20	1	19

Therefore,

$$(x, y) \in \{(10, 10), (5, 15), (4, 16), (2, 18), (1, 19)\}.$$

3. Find all positive integers x and y satisfying $xy - 2x - 3y = 10$.

Solution.

$$xy - 2x - 3y = 10 \longrightarrow xy - 2x - 3y + 6 = 16 \longrightarrow (x-3)(y-2) = 16.$$

The positive factor pairs of 16 are

$$(x-3, y-2) \longrightarrow (1, 16), (2, 8), (4, 4), (8, 2), (16, 1).$$

Therefore,

$$(x, y) \in \{(4, 18), (5, 10), (7, 6), (11, 4), (19, 3)\}.$$

4. Solve over the real numbers

$$\frac{x+1}{x-1} + \frac{x-1}{x+1} = 3.$$

Solution.

First, $x \neq \pm 1$. Then

$$\begin{aligned} \frac{x+1}{x-1} + \frac{x-1}{x+1} = 3 &\longrightarrow \frac{(x+1)(x+1) + (x-1)(x-1)}{(x-1)(x+1)} = 3 \longrightarrow \frac{(x+1)^2 + (x-1)^2}{x^2 - 1} = 3 \\ &\longrightarrow \frac{x^2 + 2x + 1 + x^2 - 2x + 1}{x^2 - 1} = 3 \longrightarrow \frac{2x^2 + 2}{x^2 - 1} = 3 \longrightarrow 2x^2 + 2 = 3(x^2 - 1) \\ &\longrightarrow 2x^2 + 2 = 3x^2 - 3 \longrightarrow x^2 = 5 \longrightarrow x = \pm\sqrt{5}. \end{aligned}$$

5. Which sets of three consecutive integers have a sum equal to their product? Prove that these are the only such sets.

Solution.

Let the three consecutive integers be $(n-1, n, n+1)$. Then

$$\begin{aligned} \longrightarrow 3n = n(n-1)(n+1) &\longrightarrow 3n = n(n^2 - 1) \longrightarrow 3n = n^3 - n \longrightarrow n^3 - 4n = 0 \\ &\longrightarrow n(n^2 - 4) = 0 \longrightarrow n(n-2)(n+2) = 0 \longrightarrow n = 0, -2, 2. \end{aligned}$$

Therefore, the corresponding triples are

$$\begin{aligned} n = -2 &\longrightarrow (-3, -2, -1), \\ n = 0 &\longrightarrow (-1, 0, 1), \\ n = 2 &\longrightarrow (1, 2, 3). \end{aligned} \quad \begin{aligned} &(n-1, n, n+1). \\ &(-3, -2, -1), (-1, 0, 1), (1, 2, 3) \end{aligned}$$

6. Determine all values of k for which $|x-1| + |x+1| = k$ has exactly two solutions.

Solution.

If $k < 2$,

The expressions change form at $x = -1$ and $x = 1$:

no solutions.

$$|x-1| + |x+1| = \begin{cases} (1-x) + (-x-1) = -2x, & x < -1, \\ (1-x) + (x+1) = 2, & -1 \leq x \leq 1, \\ (x-1) + (x+1) = 2x, & x > 1. \end{cases}$$

If $k = 2$,

$x \in [-1, 1] \longrightarrow$ infinitely many solutions.

If $k > 2$, the middle interval gives no solutions, while

Thus,

$$|x-1| + |x+1| \geq 2.$$

$$-2x = k \longrightarrow x = -\frac{k}{2},$$

$$2x = k \longrightarrow x = \frac{k}{2}.$$

Since $k > 2$, these satisfy $x < -1$ and $x > 1$, respectively.

Therefore, exactly two solutions occur when

$$k > 2.$$